

# Nikita Peganov

## AI Researcher

ML researcher working on large language models, multi-agent systems and reinforcement learning, backed by three years of production Python engineering at Yandex. Author of several research works in neuromorphic computing, RL and LLM systems. Currently exploring GenAI security and AI applications in psychology.

## Research

**LLM Code Translation, Python → Go** Yandex, 2026  
*Applied research, ML Engineer*

- built an LLM-based translator porting production Python services to Go
- designed evaluation pipeline for functional correctness of generated code

**Multi-Agent LLM Systems** 2024–2025  
*Independent research*

- benchmarked open-weight LLMs playing Mafia against each other, self-hosted on an A100 GPU
- built a multi-agent LLM strategy game to study emergent negotiation

**RL for Cloud Resource Allocation** Huawei, 2021–2022  
*Researcher, Data Scientist*

- trained an RL agent processing 4,500 requests/hour, 1.5M requests in total
- validated applicability of reinforcement learning under real production load

**Synchronization of Small-World Neuromorphic Networks** RAAI School, 2021–2022  
*Researcher*

- modeled brain-like small-world neuromorphic networks (Kuramoto model)
- studied synchronization dynamics under progressive neuron removal

## Selected Projects

**EmpathyAI — AI Psychologist Chatbot** 2025  
*Founder, Developer*

- AI psychologist chatbot with **100+ active users**; proactive check-ins and long-term analytics
- 2nd round of the “Student Startup” grant (Innovation Promotion Foundation)

**Scribo — LLM Thesis Assistant** 2024  
*Founder, Developer*

- LLM tool helping students draft course and diploma works; **1,583 users**

**frAId — AI Supervisor for Psychotherapists** 2023  
*Researcher, Developer*

- RAG-based CBT supervision bot analyzing sessions and giving feedback to psychologists

## Experience

**Yandex** Feb 2023 — Present  
*Python Developer, Infrastructure*

- develop highload services and metrics collectors for L3 infrastructure management
- built time-series forecasting models predicting service load
- designed an allocator distributing services across balancers via constrained optimization (system of inequalities, cvxpy)

## Education

**HSE University — MSc, AI Research & Entrepreneurship — GPA 7.38/10** 2024–2026

**HSE University — BSc, Software Engineering — GPA 6.29/10** 2020–2024

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## Research Interests

GenAI security (LLM-generated malware) — paper in progress

LLM agents & multi-agent systems

Reinforcement learning

AI in psychology

## Skills

### Python

PyTorch, NumPy, pandas, scikit-learn, Jupyter

### LLMs & NLP

fine-tuning, RAG, prompt engineering, evaluation, local inference

### Deep Learning

Transformers, neural networks, generative models

### Reinforcement Learning

policy optimization, reward design

### Multi-Agent Systems

LLM agents, simulation

### Applied ML

time-series forecasting, convex optimization (cvxpy)

### Infrastructure

Git, Linux, Docker, CUDA / A100

## Languages

**English** — C1

**Russian** — Native

## Achievements

Research works rated 9/10 by scientific supervisors

Winner, ITMO Open Olympiad in Computer Science (2020)

Winner, United Mathematical Olympiad (2020)

## References

Konstantin Degtiarev — Associate Professor, Software Engineering Dept., HSE

Vladimir Neverov — Head of Traffic Management Group, Yandex